Day	Group A	Group B	Topics	Session 1	Session 2	Session 3	Tools/Libraries	Objectives	Assessments
1	15/06/2026	18/06/2026	Deep Learning Foundations	Biological neuron vs artificial neuron perceptron concept components weights bias	Activation functions Sigmoid ReLU comparison use cases	Forward propagation step by step simple neural network computation	Python TensorFlow Jupyter Notebook or Google Colab Matplotlib	Build strong foundation of neural networks	
2	16/06/2026	19/06/2026	Backpropagation & Training	Loss functions MSE Cross Entropy regression vs classification loss	Gradient Descent intuition learning rate optimization challenges	Backpropagation intuition chain rule concept weight updates	TensorFlow Matplotlib Jupyter Notebook or Google Colab	Understand how models learn and optimize	
3	17/06/2026	20/06/2026	Core DL Components	Dense layers and architecture design layer stacking	Regularization Dropout overfitting vs underfitting	Batch Normalization optimizers Adam vs SGD	TensorFlow Matplotlib Jupyter Notebook or Google Colab	Learn essential building blocks of deep learning models	
4	22/06/2026	25/06/2026	Sequential Models LSTM	RNN basics sequence data examples limitations of RNN	LSTM architecture gates input forget output memory cell	Sequence modeling use cases simple LSTM implementation	TensorFlow Python Jupyter Notebook or Google Colab	Understand sequence modeling techniques	Assessment - I
5	23/06/2026	29/06/2026	NLP with BERT	Text preprocessing tokenization techniques embeddings introduction	Transformer basics attention mechanism high level	BERT for sentiment analysis pipeline implementation	TensorFlow Python Jupyter Notebook or Google Colab	Apply NLP models for text understanding	
6	24/06/2026	30/06/2026	Image Processing Fundamentals	Image representation pixels channels grayscale vs RGB	OpenCV basics image loading resizing transformations	Edge detection Canny filtering and basic operations	OpenCV Matplotlib Python Jupyter Notebook or Google Colab	Learn basic image processing techniques	
7	01-07-2026	04-07-2026	CNN Fundamentals	Convolution operation filters kernels feature extraction	Feature maps stride padding concepts	Pooling layers max pooling average pooling dimensionality reduction	TensorFlow Matplotlib Jupyter Notebook or Google Colab	Understand CNN core concepts	
8	02-07-2026	05-07-2026	CNN Model Building	CNN architecture design convolution pooling dense layers	Model training pipeline dataset preparation data augmentation	Image classification task evaluation metrics accuracy precision recall	TensorFlow OpenCV Matplotlib Jupyter Notebook or Google Colab	Build and train CNN models	
9	03-07-2026	07-07-2026	Transfer Learning & Detection	Pretrained models VGG ResNet concept of transfer learning	Fine tuning techniques freezing layers practical implementation	Object detection basics segmentation introduction U Net overview	TensorFlow OpenCV Matplotlib Jupyter Notebook or Google Colab	Use advanced computer vision techniques	Assessment - II
			End to End Project and GenAI Integration	Project pipeline data preprocessing training evaluation	Segmentation using U Net concept and workflow	Model explainability using Generative AI tools for debugging and insights	TensorFlow OpenCV Matplotlib Jupyter Notebook or Google Colab GenAI tools	Integrate all concepts into real world solution	End Semester Exam